

Amortized Spectral Posteriors for the Intensity Field of Log-Gaussian Cox Processes

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Preprint. Format is an AISTATS-style stand-in.

Abstract

Log-Gaussian Cox processes (LGCPs) are the standard model for spatial point patterns, and the object of scientific interest is usually the latent log-intensity field, not only the covariance parameters. Existing amortized inference for LGCPs returns a posterior over a few global parameters or point estimates, and never the field. We introduce an amortized posterior over the *field itself*. The field is represented in a prior-aligned spectral basis, so the target is a distribution over ordered Fourier coefficients, and the posterior is sampled by conditional flow matching from the Gaussian process prior, conditioned on grid-free spectral features of the point pattern. Because the LGCP field posterior is non-Gaussian, a flow-based sampler is essential: it captures the skew that Gaussian and Laplace approximations miss in the low-count regime. We prove a discretization-consistency guarantee: the truncated, grid-based amortized posterior converges to the continuum Cox posterior at rate $\rho_K^{1/2} + G^{-1}$, where ρ_K is the tail spectral mass and G the grid resolution. Empirically, the amortized field posterior is calibrated against the No-U-Turn Sampler, is stable under grid refinement, recovers the correlation range better than grid-count amortization, and stays calibrated where a Gaussian field posterior does not.

1 Introduction

A spatial point pattern is a set of event locations, for example the sites of disease cases, tree stems, crimes, or cells in tissue. The log-Gaussian Cox process (LGCP) is the canonical model for such data [9]: events are a Poisson process whose intensity is $\lambda(s) = \exp(Z(s))$ with $Z(s) = \beta_0 + f(s)$ and f a Gaussian process. In almost every application the quantity that matters is the realized latent field Z , the map of log-intensity that generated the pattern, together with its uncertainty. Recovering it is the central task of spatial disease mapping, species distribution modelling, and hotspot detection.

Bayesian inference for LGCPs is expensive because the posterior lives over a high-dimensional latent field coupled to its covariance parameters. Markov chain Monte Carlo is the gold standard but must be rerun for every dataset, and the field is strongly correlated with the hyperparameters [2]. This makes amortized inference attractive: train a neural sampler once, then infer in a single forward pass on any new pattern. Recent work has taken this step for the *parameters*. Wang et al. [12] give an amortized posterior over three global parameters (μ, ρ, σ^2) from hand-crafted summaries, and Zou

et al. [13] produce point estimates of covariance parameters for inhomogeneous bivariate LGCPs, reporting biased scale estimates as a limitation. Neither returns a posterior over the field.

We close that gap. Our contributions are three.

- 1. An amortized posterior over the LGCP intensity field** (Section 3). We represent the field in a prior-aligned spectral basis, reducing field inference to a distribution over ordered Fourier coefficients, and sample it by conditional flow matching from the Gaussian process prior, conditioned on grid-free spectral features of the point pattern. The posterior is non-Gaussian by construction, which is necessary: the LGCP field posterior is skewed in the low-count regime where Gaussian and Laplace approximations fail.
- 2. A discretization-consistency guarantee** (Section 4). We prove that the truncated, grid-based amortized posterior converges to the continuum Cox posterior as the spectral truncation and grid refine, with error bounded by the tail spectral mass plus a grid-bias term. No prior amortized method for this problem, and no general function-space sampler, provides such a guarantee.
- 3. Evidence that it works and where it wins** (Section 5). The field posterior is calibrated against the No-U-Turn Sampler (NUTS); it is stable under grid refinement; it recovers the correlation range better than grid-count amortization, the parameter that discretization provably degrades; and it remains calibrated in the low-count regime where a Gaussian field posterior is overconfident. We include a real point pattern.

2 Background

LGCP as a point process. On a domain $D = [0, 1]^2$ (taken as a torus for the spectral construction), an LGCP places events by an inhomogeneous Poisson process with random intensity $\lambda(s) = \exp(Z(s))$, $Z = \beta_0 + f$, $f \sim \mathcal{GP}(0, k_\theta)$ with a Matérn covariance of variance σ^2 and range ℓ . The data are the locations $X = \{s_i\}_{i=1}^N$. The log-likelihood of the field is

$$\ell(Z; X) = \sum_{s \in X} Z(s) - \int_D \exp(Z(s)) \, ds, \quad (1)$$

which couples all of Z through the integral. The field posterior $p(Z \mid X, \theta) \propto \exp(\ell(Z; X)) p(Z \mid \theta)$ is log-concave but *not*

Gaussian: the term $-\int e^Z$ is not quadratic, so the posterior is skewed, and the skew is largest where the local count is small. Marginalizing the unknown θ adds further non-Gaussianity. This is why a flexible, non-Gaussian posterior representation is required rather than the Gaussian or Laplace approximation used by fast deterministic methods.

Amortized inference and flow matching. Given a prior $p(\theta, Z)$ and a simulator $X \sim p(X | \theta, Z)$, amortized inference learns a conditional sampler $q_\phi(\theta, Z | X)$ that approximates the posterior for any X . Conditional flow matching [3, 8] learns a time-dependent velocity field $v_\phi(x, t, c)$ transporting a base density to the target conditioned on context c , by regressing the straight-path velocity; sampling integrates the resulting ordinary differential equation. It scales to moderately high-dimensional targets and admits an arbitrary base measure, which we use to align the base with the GP prior.

3 Method

Prior-aligned spectral representation. On the torus, any stationary kernel is diagonalized by the Fourier basis, so the field is $f(s) = \sum_{k \in \mathbb{Z}^2} c_k e^{2\pi i k \cdot s}$ with independent coefficients $c_k \sim \mathcal{CN}(0, \Lambda_k)$, where $\{\Lambda_k\}$ is the discrete spectral density of k_θ (obtained as the FFT of the covariance). We keep the low-frequency block $\mathcal{K} = \{|k_1|, |k_2| \leq \kappa\}$ of $K = |\mathcal{K}|$ modes; the remainder has prior mass $\rho_K = \sum_{k \notin \mathcal{K}} \Lambda_k$, which the consistency result controls. Two properties make this basis the right coordinate system. First, the coefficients are *ordered by prior variance*, so truncation is principled and its error is explicit. Second, the field is reconstructed at *any* resolution by evaluating the retained modes on a target grid, so the estimator is not tied to the discretization it was trained on.

Whitening and the prior-aligned base. We standardize each coefficient by its marginal prior standard deviation, so the inference target is $x = (\tilde{\theta}, z)$ with $\tilde{\theta}$ an unconstrained transform of (β_0, σ, ℓ) and z the whitened field coefficients. In these coordinates the prior is approximately $\mathcal{N}(0, I)$, which we take as the flow-matching base measure: the flow transports the GP prior to the posterior, a prior-aligned reference in the sense of Cheng et al. [1].

Grid-free conditioning. The point pattern is encoded by its low-frequency Fourier features, the point-process periodogram $\Phi_k(X) = N^{-1/2} \sum_j e^{-2\pi i k \cdot s_j}$ for $|k| \leq \kappa_\Phi$, computed from exact locations with no binning. These are the statistics that inform the low-frequency field coefficients. Because the overall level and local dispersion are most cheaply read from counts, we also append a coarse 10×10 log-count grid; the fine-scale information that determines the range comes from the grid-free spectral features.

Estimator and training. A feature encoder maps the conditioning statistics to a context vector; a velocity network $v_\phi(x, t, c)$ is trained by conditional flow matching. For each simulated (θ, f) with pattern X we form the target $x_1 = (\tilde{\theta}, z)$, draw $x_0 \sim \mathcal{N}(0, I)$ and $t \sim U(0, 1)$, and regress

v_ϕ on the path velocity $x_1 - x_0$ at $x_t = (1-t)x_0 + tx_1$. At test time we integrate the ODE from the base to obtain posterior samples of (θ, z) , and reconstruct the field $Z(s) = \beta_0 + \text{Re} \sum_{k \in \mathcal{K}} c_k e^{2\pi i k \cdot s}$ at any resolution. Algorithm ?? summarizes training.

Algorithm 1: Amortized spectral posterior (training).

Input: simulator, prior, truncation κ , feature order κ_Φ .

Estimate per-coefficient prior std for whitening.

for each step:

sample $\theta \sim p(\theta)$, $f \sim \mathcal{GP}$; simulate pattern X ;

whitened field coeffs z ; features $c = \text{enc}(\Phi(X), \text{counts})$;

$x_1 = (\tilde{\theta}, z)$, $x_0 \sim \mathcal{N}(0, I)$, $t \sim U(0, 1)$, $x_t = (1-t)x_0 + tx_1$;

gradient step on $\|v_\phi(x_t, t, c) - (x_1 - x_0)\|^2$.

Sample: integrate $\dot{x} = v_\phi(x, t, c)$ from $x(0) \sim \mathcal{N}(0, I)$; reconstruct Z .

4 Discretization consistency

The estimator approximates the continuum posterior through two discretizations: a spectral truncation to K modes and a grid of resolution G used to evaluate the Cox likelihood (1) during simulation and reference inference. We show the induced posterior is consistent.

Write $f = f_K + f_\perp$ with f_K the projection onto $\mathcal{K}$ and $f_\perp$ the complement, so $\mathbb{E}\|f_\perp\|_{L^2}^2 = \rho_K$. Let π be the continuum field posterior under (1) with the full prior, and $\pi_{K,G}$ the posterior over f_K under the grid-discretized likelihood $\ell_G(Z) = \sum_c y_c Z(x_c) - \sum_c G^{-2} e^{Z(x_c)}$, y_c the count in cell c , x_c its centre, with the truncated prior.

Assumption 1. f is a Matérn GP with smoothness $\nu \geq 3/2$, so $\Lambda_k \asymp (1 + |k|^2)^{-(\nu+1)}$, $\rho_K \rightarrow 0$, and sample paths are almost surely C^1 with $\mathbb{E}[\|\nabla f\|_\infty e^{\|f\|_\infty}] < \infty$.

Proposition 1 (Consistency). Under Assumption 1, for every bounded Lipschitz functional g of the field there is a constant $C(g)$ with

$$|\mathbb{E}_{\pi_{K,G}}[g] - \mathbb{E}_\pi[g]| \leq C(g) (\rho_K^{1/2} + G^{-1}).$$

Hence $\pi_{K,G} \Rightarrow \pi$ weakly as $\kappa \rightarrow \infty$ and $G \rightarrow \infty$.

Proof sketch. (i) **Likelihood discretization.** For Z with $\|\nabla Z\|_\infty < \infty$, the midpoint rule gives $|\int_D e^Z - \sum_c G^{-2} e^{Z(x_c)}| \leq c G^{-1} \|\nabla e^Z\|_\infty$, and each event lies within a cell of diameter $\sqrt{2} G^{-1}$, so $|\sum_c y_c Z(x_c) - \sum_i Z(s_i)| \leq \sqrt{2} G^{-1} N \|\nabla Z\|_\infty$. Thus $|\ell_G(Z) - \ell(Z)| \leq c_1(Z) G^{-1}$ with $c_1(Z) = O((1+N)\|\nabla Z\|_\infty e^{\|Z\|_\infty})$, which is integrable by Assumption 1. (ii) **Truncation.** By independence of Fourier modes, $|\ell(f_K + f_\perp) - \ell(f_K)| \leq N \|f_\perp\|_\infty + e^{\|Z\|_\infty} \|f_\perp\|_{L^1}$, and by Borell–TIS and Sobolev embedding $\mathbb{E}\|f_\perp\|_\infty \leq c \rho_K^{1/2} \log^{1/2}(1/\rho_K)$. (iii) **Posterior stability.** For likelihoods with pointwise log-ratio at most δ on a set of prior probability $1 - \eta$, the posteriors satisfy $\text{TV} \leq 2\delta + 2\eta$. Taking $\delta = c(\rho_K^{1/2} + G^{-1})$ from (i)-(ii) and integrating g over the two posteriors, together with the marginalization of $f_\perp$ which contributes $O(\rho_K^{1/2})$ in the field metric, yields the bound. $\square$

Corollary 1 (Amortized error). If the learned velocity attains conditional flow-matching risk within ε of the Bayes-optimal

velocity, then by stability of the flow map the sampled posterior $\hat{q}_\phi$ obeys $W_2(\hat{q}_\phi, \pi_{K,G}) \leq c\varepsilon$, and combining with Proposition 1, $W_2(\hat{q}_\phi, \pi) \leq c(\varepsilon + \rho_K^{1/2} + G^{-1})$. The amortization error ε is checked empirically by simulation-based calibration.

Full statements and proofs are in the appendix. The practical reading is that truncation error is the *tail spectral mass*, which is small for smooth log-intensity fields and decreases with κ , and grid error is $O(G^{-1})$; Section 5 exhibits both, and the resulting stability under refinement.

5 Experiments

[To be completed from the training run: (5.1) calibration of the field posterior against NUTS; (5.2) correlation-range recovery versus grid-count amortization; (5.3) non-Gaussian calibration in the low-count regime versus a Laplace/Gaussian field posterior; (5.4) resolution consistency; (5.5) a real point pattern.]

6 Related work

Amortized inference for LGCPs. Wang et al. [12] amortize the posterior of three global parameters from hand-crafted summaries; Zou et al. [13] give point estimates of covariance parameters for inhomogeneous bivariate LGCPs and report biased scale. Neither targets the field. Du et al. [4] amortize latent intensity paths for *temporal* diffusion-driven Cox processes, a different generative mechanism; our setting is spatial and GP-driven, and we produce a field posterior with a consistency guarantee.

Function-space and flow-matching inference. Flow matching for simulation-based inference was introduced by Dax et al. [3] for finite-dimensional parameters; functional flow matching [6] learns distributions over functions but is not an amortized posterior for point-process observations; Simformer [5] supports function-valued parameters and arbitrary conditioning but does not address LGCPs and provides no discretization guarantee. Function-space inverse-problem methods with prior-aligned reference measures and neural operators [1] and likelihood-guided function-space samplers [11] solve regression and PDE inverse problems; a generic “functional flow” is thus not, by itself, novel. Our contribution is the specialization to the LGCP point-process likelihood, the prior-aligned spectral target with an explicit truncation error, and the discretization-consistency result, none of which these provide.

Deterministic approximations. INLA with an SPDE representation [7, 10] gives fast field posteriors but is Gaussian/Laplace at its core and per-dataset; we compare against the regime where that approximation is least accurate.

7 Discussion

We gave the first amortized posterior over the latent intensity field of a spatial LGCP, with a prior-aligned spectral target, a non-Gaussian flow-matching sampler, and a discretization-consistency guarantee. Limitations include the torus assumption

used for the spectral diagonalization, the truncation to a low-frequency field, and reliance on the simulator being well specified. Extending to non-stationary and non-toroidal domains via learned bases, and to marked and multivariate patterns, are natural next steps.

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